Observed data:

j : group membership

Y_{ij} : outcome values, realizations of random variables

X : predictor values, treated as fixed

id	j	X_{ij}	Y_{ij}
1	A	2	5
2	A	1	4
3	B	3	2
4	B	5	5

Model specification:

$$Y_{ij} = \gamma_{00} + \gamma_{10}\big(X_{ij} - \bar{X}_{.j}\big) + \gamma_{01}\bar{X}_{.j} + u_{0j} + e_{ij}$$

Estimation

Fixed effects: $\hat{\gamma}_{00},\ \hat{\gamma}_{01},\ \hat{\gamma}_{10}$

Variance components: $\hat{\sigma}^2\ (\mathrm{Var}(e_{ij})),\ \hat{\tau}_0^2\ (\mathrm{Var}(u_{0j}))$

Method: ML / REML

Model-implied covariance of Y

$$V = \mathrm{Var}(Y \mid X, Z) = \tau^2 Z Z^\top + \sigma^2 I$$

$$V_{ij,i'j'} = \begin{cases} \tau_0^2 + \sigma^2, & i = i',\ j = j' \\ \tau_0^2, & j = j',\ i \neq i' \\ 0, & j \neq j' \end{cases}$$

Standard errors of $\hat{\boldsymbol{\gamma}}$

$$SE(\hat{\gamma}_k) = \sqrt{\mathrm{Var}(\hat{\gamma}_k)}$$

$$\mathrm{Var}(\hat{\boldsymbol{\gamma}}) = (X^\top \hat{V}^{-1} X)^{-1}$$

Hypothesis tests / confidence intervals for $\hat{\gamma}_k$

Standardized statistic: $\dfrac{\hat{\gamma}_k}{SE(\hat{\gamma}_k)}$

Choice of sampling distribution:

$\mathcal{N}(0,1)$ (large samples) or t_{df} with appropriate degrees of freedom